

Jane Doe

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SUMMARY

With around 4 years of academic and industrial experience with various applications of machine learning on different data modalities, I am hoping to pursue a doctorate in applications of technology and machine learning for social good.

PUBLICATIONS

- [2020] **Hyperlinked Publication 1 Title**
arXiv
Main contributions: Dataset compilation and annotation, experimentation with various CNN architectures, writing the complete paper
- [2019] **Hyperlinked Publication 2 Title**
Ecological Informatics
Main contributions: Reannotation of the dataset, experimentation with various background subtraction methods including CNNs and mixture models, writing the literature review and method section of the paper
- [2018] **Hyperlinked Publication n Title**
IEEE Transactions on Circuits and Systems for Video Technology
Main contributions: Entire deep learning part of the paper from literature review to programming to writing

EDUCATION

Robotics and AI, Master - 2018

The University - Islamabad, PK - **GPA:** 4.00/4.00

Thesis Hyperlinked thesis title

Achievements

- Awarded some scholarship
- Awarded some medal

Mechanical Engineering, Bachelor - 2014

The University - Islamabad, PK - **GPA:** 3.58/4.00

Thesis Non-hyperlinked final project or thesis title

ACADEMIC EXPERIENCE

Python Developer / Mar 2018 - Feb 2019

The LHC - Geneva, CH

- Prototyped, evaluated, and integrated a 3-class publication classifier. It uses title, abstract, as well as Core and Non-Core first order and second order reference fractions as input and is based on the ULMFiT text classifier
- Evaluated and integrated the improved reference matcher based on Elasticsearch queries
- Partially integrated the hierarchical clustering based author disambiguation module. The module uses features derived from author name, affiliations, and the paper title

Recurrent Neural Networks Convolutional Neural Networks Hierarchical Clustering Data Analysis Python

Graduate research student / Aug 2016 - Oct 2017

Some Lab - Islamabad, PK

- Worked on fish detection and classification in challenging underwater environments using both conventional and deep learning techniques
- Achieved state of the art 84.5 mean Average Precision on the Fish4Knowledge dataset using Gaussian Mixture Models augmented with Pixel-Wise Posteriors
- Designed and manufactured apparatus for underwater video capture in freshwater reservoirs in Pakistan for data collection

Convolutional Neural Networks Design and Manufacturing Python

Summer Student / Jun 2017 - Sep 2017

The LHC - Geneva, CH

- Configured and simulated runs of different detector-particle beam interactions
- Added a more robust track reconstruction algorithm (General Broken Lines) to the Proteus framework

C++

Teaching Assistant / Sep 2016 - May 2017

Some University - Islamabad, PK

- Helped teach these courses: X (Fall 2017), Y (Fall 2016)
- Did some other stuff

Teaching

INDUSTRIAL EXPERIENCE

Co-Founder / Oct 2019 - Mar 2021

Some Startup - Islamabad, PK

- Carried out a review of existing literature
- Carried out market research to understand the current products
- Came up with some proposition
- Designed an academically rigorous pilot study to help us validate or invalidate our above listed hypothesis
- Awarded some prize

Literature Review

Market Research

Pilot Study

AI Guy / Oct 2020 - Dec 2020

Cool Company - Islamabad, PK

- Led 4 different projects with both AI and software development components
- Came up with a pilot study to help us evaluate the concordance in our methods
- Introduced a hybrid Scrum-Kanban and agile development practices for better software development

Pilot Study

Convolutional Neural Networks

Detection

Semantic Segmentation

Python

APIs

Machine Learning Engineer / Nov 2019 - Jul 2020

Smart Cart Co - Islamabad, PK

- Researched novel approaches for large scale, yet fine grained visual classification
- Enhanced code readability and performance by redesigning and implementing it in modules

Convolutional Neural Networks

Detection

Classification

Python

C++

Design Engineer / Feb 2017 - May 2017

Design Company Co - Islamabad, PK

- Trained Convolutional Neural Networks for fine-grained vehicle detection and classification on motorways
- Achieved above 88 % accuracy over fine categories and above 95 % accuracy over toll-wise categories
- Implemented a Kalman Filter for smoothing vehicle detections, tracking, and to aid in vehicle counting

Convolutional Neural Networks

Detection

Classification

Tracking

Python

SKILLS

Programming Languages: Python, C++, C, JavaScript, Bash

Markup Languages: HTML, CSS, Markdown, $\text{\LaTeX}$

Tools: PyTorch, Keras, OpenCV, NumPy, Pandas, Flask, PyTest, Docker, MQTT, Jenkins, Terraform

Other: Literature review, Market research, Documentation, CI/CD, Version Control, Web, Data science and ML, Cloud computing, CAD

EXTRACURRICULARS

- Attended the MIT Center for Brains, Minds, and Machines (CBMM) Summer Course 2020

- Attended Neuromatch Academy 2020, a 3-week digital summer school in Computational Neuroscience, as an interactive participant
- Volunteered at the International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI) 2018
- Attended the International Conference on Machine Learning (ICML) 2018
- Attended the Human Brain Project (HBP) workshop on Neurobiology for non-specialists in 2017
- Community Mentor for the Machine Learning Foundations course by University of Washington on Coursera in 2016
- Attended the Effective Altruism Student Summit 2020
- Attended Impact Summit 2020, a 3-day conference on the intersection of technology and social good
- Attended EAGxVirtual 2020, the Effective Altruism Global digital conference

STANDARDIZED TEST SCORES

GRE | 320

Quantitative: 165

Verbal: 155

AWA: 4.0

Test Date: 20th Dec 2019

TOEFL iBT | 108

Reading: 27

Listening: 27

Speaking: 27

Writing: 27

Test Date: 25th November 2019